R22

Code No.: 22ADE12

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E / B.Tech (CSE) V Sem (Main) Examination December 2024

Big Data Analytics

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		Define in brief about the Hadoop Stack for Big Data.	(2)	1	1
2		Define about querying data in Hive.	(2)	2	2
3		Explain in brief about spark SQL.	(2)	3	2
4		Define in brief spark GraphX.	(2)	4	1
5		Explain in brief about stream processing.	(2)	5	2

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Explain in detail with diagram about the architecture of HDFS.	(5)	1	1
	(b)	Explain about the working of MapReduce. Give MapReduce example.	(5)	1	2
		(OR)			
7	(a)	Explain in detail about the block replication architecture and replication method in HDFS.	(5)	1	2
	(b)	Illustrate the Command-Line Interface of Hadoop. Write the syntax of Five commands used in Hadoop CLI.	(5)	1	3
8	(a)	Explain about Pig Latin, user defined functions and data processing operators.	(5)	2	2
	(b)	Explain about HiveQL, user defined functions and aggregate function.	(5)	2	1
		(OR)			
9	(a)	Perform comparison of traditional database and Hive.	(5)	2	3
	(b)	Explain about Pig, generating examples, and comparison with database.	(5)	2	1
10	(a)	Explain in detail about the fundamentals of Scala and functional programming.	(5)	3	2
	(b)	Define how to run spark SQL queries, tables, views, databases and select statement.	(5)	3	3
		(OR)			
11	(a)	Explain about the creation of DataFrames and DataFrames operations in spark SQL.	(5)	3	2
	(b)	Define spark concepts, resilient distributed dataset and creating RDDs.	(5)	3	1
12	(a)	Explain about designing ML system for obtaining, processing and preparing data with spark.	(5)	4	2

	(b)	Define graphs on machine learning landscape, graph structured data and PageRank.	(5)	4	2
		(OR)			
13	(a)	Explain about building a classification model with spark.	(5)	4	3
	(b)	Explain about property graph, graph operators and distributed graphs.	(5)	4	2
14	(a)	Explain in detail about spark structured streaming API and discuss a use case using spark streaming.	(5)	5	2
	(b)	Explain in detail about Apache Kafka fundamentals.	(5)	5	3
		(OR)			
15	(a)	Define about exactly-once semantics, kafka transactions and tiered storage.	(5)	5	2
	(b)	Define real-time analytics use-case of fraud detection with Kafka.	(5)	5	3

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